The dynamical locus of $\theta = 2/9$: magnitude, winding, phase

Why the selection of the Koide angle lives neither in the magnitude nor in a flat winding number, and the contrast with the gauge-protected photon anchor

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.1 Setting: which channel does $\theta = 2/9$ live in?

The Koide angle $\theta = 2/9$ is confirmed empirically to seven figures but is not derived from the geometry. It is itself not an ansatz but a *found* quantity: only the Hilbert-space translation of FFGFT (Docs. 230/231/232) and its continuation to the masses (Doc. 282) made visible that the three lepton masses are the spectrum of a hermitian Z_3 circulant on the $\mathbb{C}^3$ fibre, in whose eigenvalues $\theta = 2/9$ appears as the result of the diagonalisation. It is precisely this circulant that supplies the amplitudes a_k below. For the empirical placement of the angle – $2/9$ as the rational address of a value fixed via one reference point, not as a pole-mass exactness claim (P42) – see Doc. 292; for the dynamical locus sought here this fine distinction is immaterial, since the orbit resolution lies far above the 10^{-7} level. The preceding analysis (Doc. 290) separates, at the mass operator, a *magnitude channel* (the eigenvalues, the masses) from a *phase channel* (mixing, all-pass, holonomy), and shows that a magnitude spectrum fixes the phase only up to an all-pass factor. The question here is accordingly: in which channel does $\theta = 2/9$ sit – and what kind of quantity is it, compared with a forced anchor such as the speed of light c? The supporting computations are in `allpass_theta.py`, `allpass_carrier_selection.py` and `forced_vs_contingent.py` (numpy-only, seed 20780458).

.2 The magnitude side is counter-blind

With the signed amplitudes $a_k(\theta) = 1 + \sqrt{2}\cos(\theta + 2\pi k/3)$ $(k = 0, 1, 2)$, the symmetric Koide invariant $Q = (\sum_k a_k^2)/(\sum_k a_k)^2$ is exact and θ-*independent*. The sum $S = \sum_k a_k = 3 + \sqrt{2}\sum_k \cos(\theta + 2\pi k/3) = 3$ (three cosines spaced by 120° sum to zero), and the square sum $T = \sum_k a_k^2 = 3 + 0 + 2 \cdot \frac{3}{2} = 6$ (with $\sum_k \cos^2 = \frac{3}{2}$). Hence

$$Q(\theta) = \frac{T}{S^2} = \frac{6}{9} = \frac{2}{3} \qquad \text{for every } \theta.$$

Q ranges over $[1/3, 1]$ for positive masses; the value $2/9 = 0.2222$ lies *below* that range. So $2/9$ is no boundary, no extremum, and no fixed point of the symmetric functional – in the magnitude channel $2/9$ is entirely invisible. This is the algebraic side of the counter-blindness of the Z_3 core: the orbit members $\{1/9, 2/9, 4/9\}$ are indistinguishable to any symmetric, counting quantity.

.3 The all-pass: three channels

The locus of θ can be made visible at the all-pass. A Z_3-symmetric Blaschke product $B_\theta(z) = \prod_k (z^{-1} - \bar{a}_k)/(1 - a_k z^{-1})$ with poles $a_k = re^{i(\theta + 2\pi k/3)}$ (here $r = 0.5$) splits the phase information into three quantities, two of which are blind to θ and only one of which carries it. Numerically (for the orbit angles $\theta \in \{1/9, 2/9, 4/9\}$):

- **Magnitude** $|B_\theta| = 1$ (to $\sim 10^{-15}$) for all θ – the magnitude channel is blind.
- **Winding number** $= 3$ exactly for all θ – the *flat-topological* integer is blind. This is the numerical face of the statement that $2/(9\pi)$ is irrational and $\theta = 2/9$ is therefore not a flat topological invariant: the integer winding does not separate $2/9$ from $1/9, 4/9$.
- **Continuous phase profile** arg $B_\theta(\omega)$ – different for $1/9, 2/9, 4/9$ (at the reference point $\omega = 0$: 0.093, 0.171, 0.249; the group delay separates them too). Only the *continuous, dynamical* phase profile carries θ.

An energy/magnitude functional such as $\int |B|^2\, d\omega/2\pi$ is likewise identical for all three. The distinction of $2/9$ from $1/9, 4/9$ thus lives neither in the magnitude, nor in the integer winding, nor in any symmetric functional – only in the dynamical phase profile.

.4 An illustrative selecting mechanism

What a selector would have to do is shown by a dynamical phase reference. The three orbit members have all-pass phases $\psi = (0.093,\ 0.171,\ 0.249)$. A carrier that accumulates a holonomy Φ and phase-locks to the nearest orbit phase selects exactly one representative: $2/9$ is selected precisely when Φ lies in the $2/9$ basin $[0.132,\ 0.210]$ (width 0.078). A *random* holonomy over the orbit range selects $2/9$ with probability $\approx 1/3$ – nothing prefers the value $2/9$ without a reason.

The form of this mechanism is that of the open problem: the selection is *contingent* on an external dynamical phase (a holonomy); $2/9$ has its own basin like any member, and neither magnitude nor winding privileges it. A dynamical selector would have to supply that holonomy *forced*, from an internal reason of the supplying dynamics, not by a hand-chosen relabeling. This is the same content as the breaking condition $(\delta, \chi) \approx (0.24, \pi/2)$: χ *is* the holonomy phase, δ its free depth. (An illustration of the mechanism shape; not a derivation of the value $2/9$, and not a model of any specific carrier.)

.5 Forced or contingent: $\theta^*(\delta)$ and the photon contrast

Whether $2/9$ is *forced* like c is decided by the behaviour under variation. The $\delta = 0$ part of the breaking functional has the closed form (confirmed to 10^{-15})

$$E_3(\theta, 0) = -\tfrac{1}{2} + \tfrac{1}{\sqrt{2}} \cos 3\theta, \qquad \frac{dE_3}{d\theta}\Big|_0 = -\frac{3}{\sqrt{2}} \sin 3\theta,$$

with extrema at the Z_3 points ($\sin 3\theta = 0$). Coupling the second triplet $3'$ at strength δ and phase χ, the inner extrema move away from the Z_3 point (there are two; the lower, orbit-near branch θ^* is the one tracked). On the branch $\chi = \pi/2$ the shift is smooth and monotonic (numerical, `forced_vs_contingent.py`): θ^* rises from 0.099 ($\delta = 0.10$) through 0.223 ($\delta = 0.24$) to 0.501 ($\delta = 1.0$); $d\theta^*/d\delta$ is positive throughout (median over the δ grid 0.31, locally at $\delta^* \approx 0.24$ about 0.80). The reachable range $[0.05,\ 0.52]$ contains $2/9$ in its interior, unremarkable. The $2/9$ basin in θ space – nearest orbit member, $[1/6,\ 1/3]$, width $1/6$ – corresponds via the local slope to a δ-window of $\Delta\delta \approx 0.167/0.80 \approx 0.21$. δ is soft, not fine-tuned; $2/9$ is dragged in, it sits at no structure.

The speed of light is of a different type. In $ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2$, c is a metric coefficient, not a parameter of a functional; $ds^2 = 0$ is the light cone, and massless means $v = c$ exactly – the saturation of the inequality $v \le c$. There is no $c^*(\lambda)$; $dc/d\lambda \equiv 0$, because c is a constant of the geometry, not an extremum. This reflects the physical origin: the photon's masslessness is *gauge-symmetry protected* ($m = 0$ exactly, hence $v = c$), a *boundary*/fixed-point value. $2/9$, by contrast, is the irreducibly Z_3-*breaking* angle in the *interior* of the orbit – symmetry-breaking, not protected, and therefore contingent.

.6 Self-adjoint and skew-adjoint holonomies

If the selector is a holonomy phase χ, two sectors must be distinguished. In the breaking term $\delta \cos(2\varphi_k + \chi)$ ($\varphi_k = \theta + 2\pi k/3$), χ separates an *even* from an *odd* coupling: $\chi = 0$ (and $\chi = \pi$)

gives $\cos(2\varphi_k)$, an even, *self-adjoint*, real coupling; $\chi = \pi/2$ gives $\cos(2\varphi_k + \pi/2) = -\sin(2\varphi_k)$, an odd, *skew-adjoint*, imaginary coupling.

The two sectors have different natural values. The *self-adjoint* holonomies – flat Z_3 Wilson lines ($\chi = 2\pi n/3$) times the two spin structures per cycle (offset 0 or π) – are the multiples of $\pi/3$, $\chi \in \{0, \pi/3, 2\pi/3, \pi, 4\pi/3, 5\pi/3\}$; $\pi/2$ is none of them. The *skew-adjoint* sector, by contrast, has its natural value at $\chi = \pi/2$: the i-phase, $e^{i\pi/2} = i = \sqrt{-1}$, against the real $e^{i\pi} = -1$. The transition from a real to a skew-adjoint coupling *halves* the phase $\pi \to \pi/2$ – it takes the square root of -1.

What matters is which sector supplies the selector. The elimination chain (Doc. 282) has ruled out every static, self-adjoint candidate; the only survivor is the dynamical *skew-adjoint* phase. The type-consistent value is therefore $\chi = \pi/2$ – the i-phase of the skew-adjoint sector – not a multiple of $\pi/3$. And $\chi = \pi/2$ reaches $\theta^* = 2/9$ at $\delta \approx 0.24$ (numerically, `holonomie_landschaft.py`). The skew-adjoint type by itself admits both i-values, $\chi = \pi/2$ and $\chi = 3\pi/2$ ($e^{\pm i\pi/2} = \pm i$); the choice of branch is an orientation choice (sense of traversal of the cycle). The two branches are, however, not equivalent: the mirror branch $\chi = 3\pi/2$ has no inner extremum near $2/9$ at $\delta \approx 0.24$ (extrema at 0.83 and 1.87) – only the $+i$ branch reaches the orbit region.

$\chi = \pi/2$ is thus not an arbitrary or topologically homeless value, but the skew-adjoint i-value required by the elimination chain – with no assumed Z_4 and no numerical coincidence. Precisely, "skew-adjoint" here refers to the *generator*, not the finite holonomy: the latter is a unitary group element $\exp(K)$ with skew-adjoint K ($K^\dagger = -K$); the skew-adjoint character sits in the generating coupling, not in the finite holonomy matrix itself. The depth δ nonetheless remains *free*: the sector fixes the phase χ, not the depth. The open point is thereby sharpened – not an arbitrary angle, but whether a skew-adjoint dynamics carries precisely $\chi = \pi/2$ and thereby fixes the depth δ for an internal reason *of that dynamics* (recursion-internally, by Doc. 275, no value can be distinguished).

.7 The depth δ as the second harmonic

The mass amplitude is a Fourier series in the fibre angle,

$$a(\varphi) = \underbrace{1}_{\text{mean}} + \underbrace{\sqrt{2}\cos\varphi}_{\text{1st harmonic}} + \underbrace{\delta\cos(2\varphi + \chi)}_{\text{2nd harmonic}}, \qquad \varphi = \theta + 2\pi k/3.$$

The first harmonic, with real amplitude $c_1 = \sqrt{2}$, fixes the Koide invariant $Q = 2/3$ (it is the magnitude $r = \sqrt{2}$). The second harmonic has the complex amplitude $c_2 = \delta e^{i\chi}$; its *magnitude* is $\delta = |c_2|$ and its *phase* is $\chi = \arg c_2$. For $\chi = \pi/2$ the coefficient $c_2 = i\delta$ is purely imaginary – the second harmonic is a pure sine, the skew-adjoint coupling of the previous section.

So δ and χ are the magnitude and phase of the *same* quantity c_2, but they belong to different sectors: $\chi = \arg c_2$ lives in the phase sector (fixed to $\pi/2$ by the elimination chain), $\delta = |c_2|$ in the *magnitude* sector. The depth δ is thus not a phase but an amplitude degree of freedom – the amplitude of the first overtone, physically the strength with which the second triplet $3'$ (the doubling $6 = 3 \oplus 3'$) mixes into the primary 3.

This is the same sector that already fixes $|c_1| = \sqrt{2}$. As the first harmonic amplitude follows from $Q = 2/3$, the second, $|c_2| = \delta$, would have to follow from the overtone content of the mass-generating fractal waveform (the structure $D_f = 3 - \xi$, the $3 \oplus 3'$ doubling). The ratio $\delta/\sqrt{2} \approx 0.17$ – the second harmonic is about $17\,\%$ of the first – is a pure *shape* quantity, fixed by the waveform, not by the phase. δ is therefore not a free tuning parameter but a not-yet-computed magnitude degree of freedom.

The question of $2/9$ thereby separates cleanly into two sectors: the *phase* $\chi = \pi/2$ is skew-adjoint and fixed; the *magnitude* δ is the second harmonic amplitude and belongs – open – to the same magnitude sector as $\sqrt{2}$.

.8 δ as a golden-damped amplitude

The overtone content can be sharpened: the second harmonic amplitude is a *golden damping*. The value that gives $\theta^* = 2/9$ at $\chi = \pi/2$ is $\delta^* = 0.2389$; the golden attractor $1/\varphi^3 = 0.2361$ sits just below it. Since $1/\varphi^3 = \varphi^{-3}$ is a *negative* golden power (a decay), δ is a damping, not an amplification – $\delta^* < 1$ is unambiguously in the damping regime.

The exponent 3 is not free but $= D$. A single, decompactified $3'$ recursion would give $1/\varphi^1 = 0.618$ ($159\,\%$ off); only damping over $D = 3$ dimensions (the fractal/spatial dimension $D_f = 3 - \xi$, the *full* compact T^4) gives $1/\varphi^3$. The matching exponent 3 is thus itself an argument for the full compact T^4 and against decompactifying to a single recursion.

The ξ-damping recursion $r(k+1) = r(k)(1-\xi)$, $r(0) = D_f/3$ (Doc. 275) passes $1/\varphi$ at depth $k^* = \ln\varphi/\xi \approx 3609$ and $1/\varphi^3$ at $3 \cdot 3609 \approx 10827$ (to $0.011\,\%$). δ^* is reached about 89 *units* earlier – meaning a difference of two real-valued log-depths in the decompactified, continuous scale flow ($1/\varphi^3$ at 10826.2, δ^* at 10737.7; difference 88.5), not an integer step count. So δ^* is a $1/\varphi^3$ under-damped by about 89 log-units, an excess of $+1.187\,\%$ above the attractor – both statements are, via $\ln(\delta^*\varphi^3)/\xi$, one and the same number, not two independent findings –, consistent with an approach from above.

This excess is *not* the fractal mass correction $K_{\text{frak}} = 1 - 100\xi$ (Doc. 133): its precise form $(D_f^{\text{eff}}/3)^{D_f^{\text{eff}}/2}$ with $D_f^{\text{eff}} \approx 2.973$ gives a factor ≈ 100 ($1.333\,\%$), whereas δ needs only $\approx 89\,\xi$ ($1.187\,\%$) – less, hence not K_{frak}. And the recursion depth is explicitly *not* independently fixed in Doc. 275: there $1/\varphi^3$ is a *transit point*, not a fixed point (the only fixed point of the damping is 0), and for *every* target $c \in (0,1)$ a depth $k^*(c)$ exists with identical parameter status. That δ^* lies near $1/\varphi^3$ is therefore *not* a closing of the recursion, and fixing the depth to that value would, by Doc. 275, be no distinction. The 89-step excess therefore remains an open, underived fine value.

So the magnitude sector is clarified as far as the structure carries: δ is the golden-damped second harmonic, $1/\varphi^3$ over $D = 3$ (full T^4), with a small excess from above. Form, order, exponent and direction are fixed; the exact δ value – the 89-step offset – is neither K_{frak} nor a fixed recursion depth and remains open (`delta_golden_daempfung.py`).

.9 What the non-closure means in practice: an open coupling channel

The picture so far separates two sectors with opposite behaviour. The mass sector is *closed*: the recursion $R(\Phi^*) = \Phi^*$ (Doc. 203) has a non-trivial fixed point, and because the lepton masses sit on it, they and the Koide value $Q = 2/3$ are exact and rigid.[1] There is nothing to adjust there – no free parameter, no point of entry for an external influence.

The room sits solely in the open sector, the phase. The ξ-damping recursion never closes (transit point, only fixed point 0, Doc. 275); in feedback language it is an open, damped loop with gain below one, not a rigidly locked fixed point. Such an open, damped loop can – unlike a closed fixed point – be influenced from outside: by coupling to a second oscillator, by injection or mode-locking (Doc. 286). In practice the non-closure therefore means that a channel exists through which a second, dynamical system can lock the open phase. The closed mass sector offers no such channel.

[1] "Exact" here refers to the geometric fixed-point level: the dimensionless invariant $Q = 2/3$ is rigid and is confirmed by the data to $-0.9\,\sigma$ (Doc. 292). The *absolute* masses, or the number pair $(\sqrt{2},\ 2/9)$, must be distinguished from this: on the measured pole-mass level the pair is not jointly exact (Doc. 292, $\sim 450\,\sigma$ in the m_μ/m_e channel), which is why $2/9$ is treated per P42 as the rational address of an angle fixed via one reference point, not as a pole-mass exactness claim. For the dynamical locus discussed here this is immaterial – the orbit resolution (~ 0.1) lies far above the 10^{-7} difference.

The channel is, however, narrow and structured, not a free parameter. The capture range of such a coupling is of order ξ (Doc. 286) – tiny; the $2/9$ Arnold tongue is about twelve times narrower than $1/3$. The coupling of adjacent scale levels in the continuous flow is real but of order 10^{-13} in magnitude and does not destabilise the closed predictions. And the effect is directed: it runs from the dynamics into the phase, not into the radial mass sector, which stays rigid (the radial/angular split of Doc. 282).

The one place this channel would be used is a dynamical skew-adjoint generator outside the static FFGFT geometry – the continuation of the elimination chain (Doc. 283/286). If such a generator locks the open phase precisely to $\chi = \pi/2$ and thereby $\theta = 2/9$, then $2/9$ would acquire the dynamical, protected status the following conclusion describes. This is explicitly a candidate, not an attained result: FFGFT's own recursion R is purely real and carries no such generator; whether an external dynamics supplies one remains open.

The decisive point is the boundary of this room. "Room for mutual influence" does not mean the theory can be bent at will. The room is bounded: the radial sector stays rigid, the phase is tightly constrained (skew-adjoint, $\chi = \pi/2$, transcendental in 2π, ξ-narrow capture range), and the recursion depth fixes no value (Doc. 275). It is structured room, not a free parameter – and it is exactly this boundary that distinguishes a channel for genuine coupling from an after-the-fact numerical adjustment.

This channel is also the reason the fundamental sector does not collapse to a rigid skeleton. A fully closed model – every value a forced fixed point – would be over-determined: no contingent selection within a Z_3 orbit, no flavour mixing, no physical (non-removable) phase. Mixing in particular is phase, not magnitude (Doc. 289: charged leptons magnitude-dominated with small mixing, neutrinos phase-dominated with large mixing, the pure all-pass sector). The sharpest case is the structure of CP violation: it requires an irreducible complex phase – a purely real, self-adjoint structure is CP-conserving – hence exactly the type of the skew-adjoint channel $\chi = \pi/2$. The open phase channel is thus structurally of the kind that can carry CP violation; a fully self-adjoint FFGFT would be CP-even and non-mixing. The scope stays narrow: the channel carries mixing, CP and contingency in the *fundamental sector* – not the macroscopic diversity (chemistry, structure formation), which has its own independent sources.

.10 Conclusion: the open dynamical question

Statically the situation is clear: $2/9$ is invisible in the symmetric Q, blind to magnitude and to the integer winding, and an interior extremum dragged in by a free parameter ($d\theta^*/d\delta \neq 0$). It is *not* forced like c. Both quantities share the same flat T^4 arena – c as a signature boundary, $2/9$ as an interior phase degree of freedom – but they are of different types.

For $2/9$ to acquire a c-like, forced status, it would need a *dynamical* fixed point, as gauge symmetry forces $m = 0$ (and hence $v = c$). The preceding sections localise this fixed point: the *phase* $\chi = \pi/2$ is the skew-adjoint i-value required by the elimination chain – the selector is type-fixed; what remains open is only the *magnitude* δ (the golden-damped overtone amplitude). Its fine value is, moreover, not "not yet" derived but *provably not* derivable from the recursion alone: the non-closure is a theorem (the only fixed point is 0), and the resonance density of the transit points also rules out any recursion-internal distinction of a value (Doc. 275, epistemic limit). The recursion side is thereby settled, not an open question; what is open is solely the existence of the external anchor – whether a skew-adjoint dynamics carries precisely $\chi = \pi/2$ and thereby fixes the depth δ for an internal reason *of that dynamics* (not of the recursion); if it did, $2/9$ would acquire a dynamical, protected privilege, otherwise it stays contingent. The elimination chain (Doc. 282) has ruled out every static candidate – symmetric, counting-topological, radial, static-geometric – and the only survivor is the dynamical skew-adjoint phase, whose origin lies outside the static geometry. It is there, not in the magnitude and not in a flat winding number, that the value $2/9$ would have to be sought.

.11 Placement: usefulness and proximity to application

The findings of this document are structure and exclusion results, not technology. Their immediate usefulness lies with three groups. First, work on the Koide relation and flavour structure: the no-go ($2/(9\pi)$ irrational; magnitude and integer winding counter-blind) closes off topological and counting derivation routes for $\theta = 2/9$ and spares their further pursuit; the elimination chain localizes the remaining search precisely in the dynamical skew-adjoint phase channel. Second, methodologically: the magnitude/phase decomposition serves as a shared audit language for framework comparisons (script-backed claims, pre-registered criteria, matched nulls) and transfers independently of the FFGFT content. Third, didactically: the translation between particle physics and communications engineering (circulant as DFT, all-pass, injection locking, Arnold tongues) makes the phase channel computable with standard engineering tools.

Only the phase channel itself touches measurement: the structural statement that mixing and CP violation live in the skew-adjoint channel (Doc. 289) has a falsifiable form in the oscillation phases and δ_{CP}, which running and upcoming experiments (DUNE, Hyper-Kamiokande) measure. An application in the narrow sense – a device, a procedure, short-term engineering use – does not follow: the tools run from communications engineering into physics, not back, and as long as δ is not derived, the yield is that of a sharpened open problem, not of a result.